

Yuxiang Liu

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EDUCATION

University of California Berkeley

Bachelor of Arts in Computer Science - GPA 3.942/4.0

Expected May 2027

Relevant Coursework: Probabilistic Model in Computer Science (EECS 126 A) **A**, Optimization Models in Engineering (EECS 127) **A+**, Model Predictive Control (ME 231A) **A**, Robotics Dynamics (EECS 106A) **A**, Feedback Control (EE 128) **A+**, Mathematical Analysis (MATH 104) **A+**

ACADEMIC EXPERIENCES

Research Intern, Model Predictive Control Lab, UC Berkeley

Jan. 2024 – Present

- Enabled sim-to-real of a PPO-trained wheeled quadruped by incorporating realistic joint-actuator delays into training.
- Developed BARC-CARLA, a simulation environment for end-to-end racing based on CARLA.
- Developed LTrack-Go2W, a Mujoco-based parallel simulation environment for wheeled-quadruped racing.

Summer Research Intern, AMBER Lab, California Institute of Technology

Jun. 2026 – Aug. 2026

- Reproduced fast-UMI data collection pipeline, using CAD modeling to customize for the handshake of Unitree-G1.

Officially Invited Reviewer and Oral Presenter of 23rd IFAC World Congress

FIRST-AUTHOR PUBLICATIONS

State-Conditional Adversarial Learning: An Off-Policy Visual Domain Transfer Method for End-to-End Imitation Learning

[Yuxiang Liu, Shengfan Cao]

Accepted, IFAC 2026

- Derived a target-domain imitation-loss bound linking source-domain error to state-conditional latent KL divergence, establishing a tractable objective for visual domain transfer without explicit expert supervision.
- Developed SCAIL, an adversarial-learning framework that aligns state-conditioned latent distributions using off-policy target data without target-domain expert supervision.
- Exceeded expert-supervised DAgger trajectory lengths across three target sampling distributions with as few as 256 supervision-free samples; demonstrated low-to-high-speed transfer in the task of end-to-end vision-based racing car.

The Layer Mystery of VLA: An Information-Theoretical Analysis of VLA Latent Interface

[Yuxiang Liu, Lizhi Yang, Fengze Xie, Aaron Ames, Yisong Yue]

Under review, ICLR 2027

- Compared 54 fusion configurations of VLA latent interface across three frozen VLM backbones and two benchmarks (LIBERO, CALVIN), with three seeds each; 47 underperformed the best observed single-layer policy.
- Developed InfoNCE-based layer selection requiring 9–33× less GPU compute than exhaustive sweeps and reducing mean selection regret from 17.89 to 3.71 percentage points versus deepest-layer selection.
- Derived a reweighting equivalence between information-bottleneck objectives for action prediction and action-conditioned InfoNCE, providing a theoretical basis for efficient layer selection.

RACER: Residual-Adaptive Closed-Loop Estimation for Sampling-Based Planning in Wheeled-Quadruped Racing

[Yuxiang Liu, Marla Eisman, Lizhi Yang, Aaron Ames, Francesco Borrelli]

Under review, ICRA 2027

- Developed RACER, a high-speed wheeled-quadruped racing control stack combining sampling-based MPC, learned residual dynamics, and an RL velocity tracker.
- Proposed Low-Rank Residual Adaptation (LoRRA), transferring dynamics pretrained on 10 million simulated transitions using ~400 hardware transitions while updating only 17.7% of parameters.
- Our policy able to complete all 9 hardware laps with average speed of 3.2 m/s, averaging 1.95 m/s, matching the performance of human-teleoperator.

TECHNICAL SKILLS

- **Development Tools:** Mujoco, Isaac Lab, MATLAB / Simulink, ROS2, Docker, Ubuntu System
- **Core Expertise:** GAN, Diffusion Model, Flow Matching, LoRA, MPPI, DIAL-MPC, Sim-to-Real Transfer of RL Policy, PPO Training, Action-expert / Action-head Training of Vision-Language-Action Models, InfoNCE